

The Price-Tag Decision: Retail Relabel Assignment
ISM6642 – Machine Learning for Business Applications | Dr. Hemang Subramanian
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Part 1 — Feature Engineering and EDA

We started by checking the data (1,500 SKUs, 13 columns, no missing values) and creating five new variables that help explain why the owner changes a price tag. We calculated the average price of the three competitors and used it to calculate the percentage gap between the store price and the competitor average, the absolute percentage gap, the log of weekly units sold, and the margin percentage.

Exhibit 1. Engineered features, summary statistics, and hand check on SKU10000

Feature	Mean	Std. dev.	Median	Formula	SKU10000 check
CompAvg	19.35	25.38	10.32	$(\text{BigBox} + \text{ValueMart} + \text{QuickShop}) / 3$	$(2.46 + 3.34 + 2.22) / 3 = 2.673$
GapPct	3.62	12.73	3.71	$100 \times (\text{StorePrice} - \text{CompAvg}) / \text{CompAvg}$	$100 \times (3.06 - 2.673) / 2.673 = 14.46$
AbsGapPct	11.00	7.36	10.10	$ \text{GapPct} $	14.46
LogUnits	3.94	0.50	3.99	$\ln(\text{WeeklyUnitsSold})$	$\ln(71) = 4.263$
MarginPct	32.68	7.74	33.74	$100 \times (\text{StorePrice} - \text{UnitCost}) / \text{StorePrice}$	$100 \times (3.06 - 1.74) / 3.06 = 43.14$

Source: Team's Google Colab Notebook, Steps 0-1.

The average signed gap was 3.62%, while the average absolute gap was 11.00%. The difference is large because positive and negative gaps cancel each other out when they are averaged: a product 10% above competitors and one 10% below average to 0%, even though both are 10% away. The absolute gap keeps the size of the misalignment. Most products (883 of 1,500, or 59%) are priced above the competitor average. Overall, 40.0% of products were relabeled. Category rates were similar, ranging from 37.8% for Household to 44.0% for Electronics. Paper-tag products had a 33.9% relabel rate and an average relabel cost of \$1.45, while electronic shelf labels had a 54.1% rate and an average cost of \$0.06.

Exhibit 2. Relabel base rates: overall, by category, and by shelf-label type

Group	Relabel rate	SKUs	Avg. RelabelCost
Overall	40.0%	1,500	\$1.03
Electronics	44.0%	191	-
Grocery	41.2%	486	-
Apparel	39.0%	287	-
PersonalCare	38.3%	240	-
Household	37.8%	296	-
Paper tag (ESL = 0)	33.9%	1,045	\$1.45
Electronic shelf label (ESL = 1)	54.1%	455	\$0.06

Source: Team's Google Colab Notebook, Step 1.

This is an example of a menu cost, which is the cost of physically changing a price. When the tag is expensive to change (paper, \$1.45), the store may leave a small price error alone because fixing it is not worth the effort. When changing the tag is almost free (ESL, \$0.06), the store can fix even small errors.

Part 2 — A Naive Model

We used logistic regression because Relabel has two possible outcomes: 0 means the product was not relabeled and 1 means it was relabeled. Our first model used GapPct, UnitCost, and RelabelCost as

predictors. The model produced a McFadden pseudo R-squared of 0.077 and an AIC of 1871.0. GapPct had a positive and statistically significant coefficient, UnitCost was not statistically significant, and RelabelCost was negative and statistically significant.

Exhibit 3. Model 1: Relabel ~ GapPct + UnitCost + RelabelCost

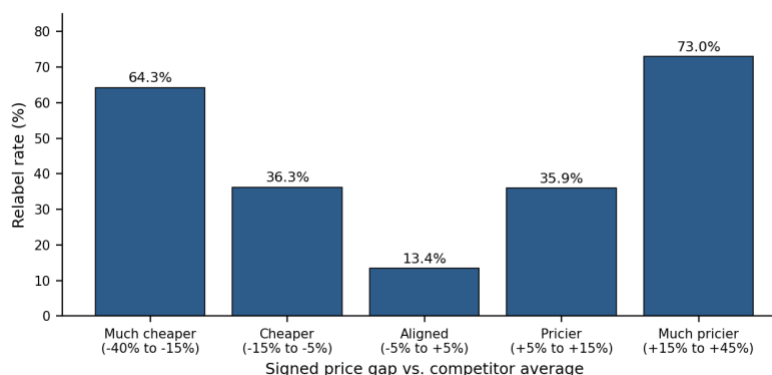
Variable	Coefficient	Std. err.	p-value	Odds ratio	95% CI (odds ratio)
Intercept	0.0589	0.0977	0.547	1.061	0.876 – 1.285
GapPct	0.0379	0.0045	< 0.001	1.039	1.030 – 1.048
UnitCost	0.0005	0.0032	0.877	1.000	0.994 – 1.007
RelabelCost	-0.6387	0.0707	< 0.001	0.528	0.460 – 0.607

Model fit: pseudo $R^2 = 0.077$; AIC = 1871.0; BIC = 1892.3; $n = 1,500$. Odds ratio = $\exp(\text{coefficient})$.

Source: Team's Google Colab Notebook, Step 2.

However, we found a problem with using the signed GapPct as the main price-misalignment variable. A signed gap treats being above competitors differently from being below competitors, even though both situations can create a reason to change the price. For example, a product priced 10% below competitors may be losing potential margin, while a product priced 10% above competitors may be losing customers. In both cases, the price is 10% away from the competitive price.

Exhibit 4. Relabel rate by signed price gap



SKUs per group, left to right: 98, 331, 380, 373, 318. Source: Team's Google Colab Notebook, Step 1.

The problem is that a logistic model with one GapPct term can only draw a line that goes in one direction. Because the coefficient is positive, the model says that the cheaper a product is, the less likely it is to be relabeled. This means it treats a product 10% below competitors as less likely to be relabeled than a product that is perfectly aligned. That is the opposite of what the data show (Exhibit 4): products far below competitors were relabeled 64.3% of the time, compared with only 13.4% for products priced within $\pm 5\%$ of competitors. GapPct is still significant only because more products are priced above competitors (318 in the top group vs. 98 in the bottom group), so the line fits the expensive side and misses the cheap side. The effect is diluted, not completely hidden, which is also why the pseudo R^2 is only 0.077. Therefore, what matters to the owner is how far a price is from competitors, not the direction. The relationship is U-shaped, so AbsGapPct is the correct functional form (the right way to put the price gap into the model), and Part 3 uses it in Model 2.

Part 3 — The Economically Motivated Model

For Model 2, we replaced GapPct with AbsGapPct, which measures how far the price is from competitors without caring about the direction. We also added three variables the business scenario points to: CostChangePct (pressure from supplier cost changes), LogUnits (how fast the product sells), and DaysSinceLastChange (how old the current price is). RelabelCost and UnitCost stay in the model.

$$\text{Relabel} \sim \text{AbsGapPct} + \text{CostChangePct} + \text{LogUnits} + \text{RelabelCost} + \text{DaysSinceLastChange} + \text{UnitCost}$$

Exhibit 5. Model 2 Results

Variable	Coefficient	Std. err.	p-value	Odds ratio	95% CI (odds ratio)
Intercept	-4.2772	0.5673	< 0.001	0.014	0.005 – 0.042
AbsGapPct	0.1778	0.0109	< 0.001	1.195	1.169 – 1.220
CostChangePct	0.1276	0.0168	< 0.001	1.136	1.099 – 1.174
LogUnits	0.4324	0.1311	< 0.001	1.541	1.192 – 1.992
RelabelCost	-0.8589	0.0852	< 0.001	0.424	0.359 – 0.501
DaysSinceLastChange	0.0040	0.0006	< 0.001	1.004	1.003 – 1.005
UnitCost	0.0004	0.0037	0.919	1.000	0.993 – 1.008

Model fit: $\text{pseudo } R^2 = 0.275$; $AIC = 1477.8$; $BIC = 1515.0$; $n = 1,500$. Source: Team's Google Colab Notebook, Step 3.

An odds ratio (OR) above 1 raises the odds of relabeling, and one below 1 lowers them. For five of the six variables, the 95% confidence interval does not include 1, so these effects are statistically significant at the 5% level and their direction is clear.

AbsGapPct (OR = 1.195): Each 1 percentage point increase in the size of the price gap vs. competitors, in either direction, raises the odds of a relabel by about 19.5%, holding everything else constant.

CostChangePct (OR = 1.136): Each 1 percentage point increase in the supplier's cost change raises the odds of relabeling by about 13.6%.

LogUnits (OR = 1.541): Since this is the log of units sold, a 1-unit increase (sales about 2.7 times higher) raises the odds of relabeling by 54%. Put simply, when weekly sales double, the odds go up by about 35%. Faster-selling SKUs get corrected more.

RelabelCost (OR = 0.424): Each additional \$1 it costs to change the tag cuts the odds of relabeling by about 58%. This is the core menu-cost result: items that are expensive to relabel get left misaligned.

DaysSinceLastChange (OR = 1.004): Each extra day since the last price change raises the odds by 0.4%, a small but real "overdue for a correction" effect. Over 100 days, it adds up to about 49% higher odds.

UnitCost (OR = 1.000) is the exception. Its confidence interval (0.993 – 1.008) includes 1, which matches its p-value of 0.919, so we cannot rule out zero effect. CostChangePct, on the other hand, is significant. This makes business sense: a product with a high cost does not automatically need a new price, because its current price was set knowing that cost. What matters is whether the cost has changed since then. A product that has cost \$50 for a long time gives no reason to change its tag, but if the supplier suddenly raises the cost, the old price no longer covers it the same way. So a change in cost matters more than the cost level.

Exhibit 6. Model 1 versus Model 2

Model	Pseudo R^2	AIC	BIC
Model 1 — signed gap (3 predictors)	0.077	1871.0	1892.3
Model 2 — economic (6 predictors)	0.275	1477.8	1515.0

Higher pseudo R^2 is better; lower AIC and BIC are better. Source: Team's Google Colab Notebook, Step 3.

Model 2 is a much better fit: its pseudo R^2 is more than 3 times higher, and AIC and BIC both drop by almost 400 points (a drop of more than 10 is already strong evidence). Because BIC gives a bigger penalty for extra variables, the better fit is not just because Model 2 has more variables. Two things drive this jump. First, the form of the gap: when we put the signed gap back into Model 2 and change nothing else, pseudo R^2 falls from 0.275 to 0.125 (team notebook, Step 6), so using the absolute gap explains a big part of the improvement. This makes economic sense, because a store relabels items that are overpriced or underpriced,

not just overpriced ones. Second, the added variables (CostChangePct, LogUnits, and DaysSinceLastChange) are all significant and capture real reasons for relabeling that Model 1 leaves out. In both models, UnitCost is never significant, while RelabelCost is significant and negative: the cost of changing the price matters much more than the cost of the item. For these reasons, we would deploy Model 2.

Part 4 — Predictive Evaluation

We split the data into 70% for training (1,050 SKUs) and 30% for testing (450 SKUs). The split was stratified, so both groups keep the same 40% relabel rate, and we used `random_state = 7` as required. We refit Model 2 on the training data only and then tested it on the 450 products it had never seen, using a 0.5 cutoff.

Exhibit 7. Test-set confusion matrix (450 SKUs, 0.5 cutoff)

Confusion Matrix	Predicted: no relabel	Predicted: relabel
Actual: no relabel	233 (correct)	37 (false alarm)
Actual: relabel	64 (missed)	116 (correct)

Test accuracy = 0.776; test AUC = 0.853; baseline accuracy (always predict “no relabel”) = 0.600.

Source: Team’s Google Colab Notebook, Step 4.

On the 450 test products the model had never seen, it classified 77.6% correctly, well above the 60% we would get by always predicting “no relabel.” The test AUC was 0.853: if we pick one product that was relabeled and one that was not, the model gives the relabeled one a higher probability about 85% of the time (0.5 would be a coin flip). The test results are as good as or slightly better than the training results (accuracy 76.5%, AUC 0.821), so there is no sign of overfitting, and we can trust the Model 2 findings to hold for new products.

The model’s mistakes are not balanced. At the 0.5 cutoff, it misses about 36% of real relabels (64 of 180) but wrongly flags only 14% of products that were left alone (37 of 270). Because only 40% of products are relabeled, the model needs strong evidence before it predicts a relabel. Part 5 suggests that missing a needed price change costs more than a false alarm, so a cutoff lower than 0.5 would be worth testing: it would catch more real relabels, with a few more false alarms. The model still gets about 22% of products wrong, which suggests that other things not in the data, such as promotions or a manager’s judgment, may also affect relabeling. So it is best used to rank and prioritize products, not as a strict rule for which ones must be relabeled.

Part 5 — Economic Decision Analysis

For each test product, we estimated the weekly loss from keeping a misaligned price and compared it with the cost of changing the tag:

$$\text{ExpectedWeeklyLoss} = (\text{AbsGapPct} / 100) \times \text{StorePrice} \times \text{WeeklyUnitsSold}$$

$$\text{NetBenefit} = \text{ExpectedWeeklyLoss} - \text{RelabelCost}$$

A positive NetBenefit means that one week of the loss is larger than the cost of a new tag. 444 of the 450 test products (98.7%) had a positive NetBenefit, so almost all of them would recover the cost of relabeling within one week.

This number should be read carefully. The formula assumes that every unit sold loses the full price gap, which is not realistic. An overpriced product loses some customers, not all of its sales, and an underpriced product loses margin, not revenue. So ExpectedWeeklyLoss shows the most the store could be losing (an upper bound), and it is most useful for ranking which products to fix first, not as an exact dollar amount.

Still, the result is striking: almost every product seems worth fixing, but the store only relabeled 40% of them.

Exhibit 8. Median weekly dollars at stake by segment (test set)

Segment	Median expected weekly loss	Median net benefit
<i>By size of price gap</i>		
0–5%	\$11.50	\$10.41
5–15%	\$54.88	\$53.40
15% or more	\$125.79	\$124.98
<i>By sales speed (weekly units sold)</i>		
Slowest 25%	\$23.37	\$21.83
Fastest 25%	\$103.61	\$103.54
<i>By category (highest and lowest)</i>		
Electronics	\$284.38	\$284.34
Apparel	\$112.86	\$110.52
Grocery	\$21.62	\$20.62

Source: Team's Google Colab Notebook, Step 5.

Bigger gaps, faster sellers, and Electronics and Apparel products carry the largest weekly losses. The memo below turns these results into recommendations for the owner.

Memo to the Store Owner

TO: Store Owner, SunCoast

FROM: Pricing Analytics Team (Priscila, Liliette, Yasori)

DATE: October 2, 2026

SUBJECT: Which price tags to fix first, and whether to add more electronic labels

Almost every product with a price different from competitors is worth fixing. In our test group, 444 of 450 products (98.7%) would recover the cost of a new tag within one week, but last cycle the store relabeled only 40% of products.

(a) What to relabel first. Start with products whose price is 15% or more away from competitors, in either direction. Their typical (median) weekly loss is \$125.79, compared with \$11.50 for products within 5%. Next, focus on fast sellers: the fastest 25% lose a median of \$103.61 per week vs. \$23.37 for the slowest. By category, Electronics and Apparel lose the most. Products with a recent supplier cost increase, and products whose price has not changed in a long time, should also be reviewed.

(b) Electronic shelf labels. Changing a paper tag costs about \$1.45, and changing an electronic label costs about \$0.06, a saving of about \$1.39 per change. Our model estimates that if paper-tag products were as cheap to relabel as products with electronic labels, their relabel rate would rise from about 34% to 54%, so more pricing errors would get fixed. To decide if the investment pays off, we still need the cost of buying and installing the labels.

(c) Caution. These results come from one cycle of store data, not from an experiment. The electronic labels were not placed on random shelves; for example, they may have been installed on busier aisles. So the results show relationships, not proof that electronic labels cause more relabeling. A small pilot, converting some randomly chosen paper shelves and comparing them with the rest, would give stronger evidence.